MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Aeronautical Engineering/Automobile Engineering**	**Course Type**	**C/FC/PE/OE**
Course Name	**Optimisation Techniques**	**Course Code**	**AAE 4062**
Semester	**VII**	**Max Marks**	**50**
Date	**25-11-2026**	**Duration**	**3 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A.	Compute three iterations of Golden section method to maximise the function $y = 12x - x^2$ for the original interval of uncertainty of $0 \leq x \leq 10$	***(02)***	3	1	3	**C1**	**2**
1B.	A company can advertise the product by using local radio and T.V. stations. Its budget limits the advertisement expenditure to Rs. 10,000 per month. Each minute of radio advertisement costs Rs. 50 and each minute of T.V. advertisement costs Rs. 1,000. The company would like to use the radio twice as much as the T.V. Past experience shows that each minute of T.V. advertisement will usually generate 25 times as many sales as each minute of radio advertisement. Formulate this problem as an Optimisation model to maximse the sales.	***(03)***	1	4	1	**C2**	**3**
1C.	Compare the interval halving method and Fibonacci method in terms of obtained interval after 5 function evaluation of the function $e^x - x^3$ in the interval (2,5). How does the outcome change if the interval (-2,5) is chosen?	***(05)***	3	4	3	**C3**	**5**
2A.	Find minimum of the function f=(x-1)(x-2)(x-3) by Newton's method in the range (1,3) Compute three iterations	***(02)***	3	3	3	**C1**	**3**
2B.	Locate and classify stationary points for the function $f(x_1, x_2) = 10(x_2 - x_1^2)^2 + (1 - x_1)^2$	***(02)***	2	2	2	**C1**	**4**
2C.	Compute two iterations of simplex method to find minimum of the function $10 - x_1 + x_1x_2 + x_2^2$with initial simplex as [0, 2]^T, [0, 0]^T, [1, 1]^T. Assume Reflection, expansion and contraction coefficient as 1.0, 2.0 and 0.5 respectively.	***(05)***	3	3	3	**C2**	**3**

<table>
<tr><td>3A.</td><td>What are the differences between the Cauchy's and Newton's search methods.</td><td>(02)</td><td>3</td><td>1</td><td>3</td><td>C1</td><td>2</td></tr>
<tr><td>3B.</td><td>For a vector $s_1 = \left[\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]^T$ find direction s_2 conjugate to s_1 with respect to the objective function $f(x) = x_1 + 2x_2^2 - x_1x_2 + 3x_1^2 - 2x_1x_3 + x_3^2$</td><td>(03)</td><td>2</td><td>2</td><td>2</td><td>C1</td><td>4</td></tr>
<tr><td>3C.</td><td>Perform two iterations of the steepest descent method to minimize the function $f(x_1, x_2) = 100(x_2 - x_1)^2 + (1 - x_1)^2$ from the starting point $[-1.2, -1.0]^T$</td><td>(05)</td><td>3</td><td>3</td><td>3</td><td>C2</td><td>3</td></tr>
<tr><td>4A.</td><td>Find the shortest distance from the point [3,0] to the parabola $y=x^2$ using Lagrange multiplier method</td><td>(02)</td><td>3</td><td>3</td><td>3</td><td>C2</td><td>4</td></tr>
<tr><td>4B.</td><td>Use three itrations of the secant method to find the minimum of the function $f(x) = e^{0.2x} - (x+3)^2 - 0.01x^4$</td><td>(03)</td><td>2</td><td>1</td><td>2</td><td>C2</td><td>2</td></tr>
<tr><td>4C.</td><td>A circular tank that is closed at both ends is to be fabricated to have a volume of 250π m^3. The fabrication cost is found to be proportional to the surface area of the sheet metal needed for fabrication of the tank and is Rs.400/m^2. The tank is to be housed in a shed with a sloping roof which limits the height of the tank by the relation $H \leq 8D$, where H is the height and D is the diameter of the tank.
Write KKT necessary conditions.
Solve KKT necessary conditions for local minimum points. Check sufficiency conditions</td><td>(05)</td><td>3</td><td>4</td><td>3</td><td>C3</td><td>3</td></tr>
<tr><td>5A.</td><td>The population of binary strings in a maximization problem is given below:
<table>
<tr><th colspan="6">String</th><th>Fitness</th></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>8</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>12</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>6</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>2</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>18</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>9</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>10</td></tr>
</table>
Determine the expected number of copies of the best string in the above population in the mating pool using the roulette-wheel selection process</td><td>(02)</td><td>4</td><td>2</td><td>4</td><td>C2</td><td>3</td></tr>
<tr><td>5B.</td><td>List and describe genetic operators.</td><td>(03)</td><td>4</td><td>1</td><td>4</td><td>C1</td><td>2</td></tr>
</table>

5C.	Perform two iterations of the Newton's method to minimize the function $f(x_1, x_2) = 100(x_2 - x_1)^2 + (1 - x_1)^2$ from the starting point $[-1.2, -1.0]^T$	***(05)***	3	3	3	**C2**	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	**3**
2	**8**
3	**33**
4	**5**

BL	Total Marks
2	**11**
3	**27**
4	**7**
5	**5**